

Certified analytical bounds for hypersonic reachable sets

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Abstract

A hypersonic entry vehicle may cross the dense part of an atmosphere several times in one descent, and each crossing occupies an interval of time that vanishes as the atmosphere is made thin. Whether the effect of a crossing persists in that limit, and in which variables it persists, depends on a distinguished scaling of the entry dynamics which has to be identified rather than assumed; settling it, and determining whether the resulting family admits a limit that retains the crossings, is the first question of this proposal. What I want from the work is a rigorous and quantitative guarantee for atmospheric entry: explicit outer bounds on the reachable set, uniform over the admissible guidance and over an atmosphere known only within bounds, together with the hypotheses such a bound requires. The structure I expect to make that tractable is the one familiar from concentration compactness — a skeleton carrying the slow descent, one profile for each crossing, and a remainder — and it is a hypothesis to be earned rather than a result. My training for it is joint work with E. Kirr on multi-peak ground states of the nonlinear Schrödinger equation, where a concentration-compactness decomposition and a Lyapunov–Schmidt reduction together classify the configurations of concentration points. This document poses the problem, gives a reduced model and the questions that must be settled before it can be justified, states what would have to be established, and sets out a plan of work; nothing in it is claimed as proved.

1 The question

What I want is a rigorous and quantitative guarantee for atmospheric entry: an outer bound on the states a vehicle can be driven to, holding for every admissible control and for every atmosphere within stated bounds. What I expect to be reachable is the strongest such guarantee the full dynamics support, stated with the hypotheses it requires, and not an unconditional bound: some effects will need stronger assumptions than others and some may obstruct a guarantee altogether, and saying which is part of the result. That is the objective of the thesis and the reason for the work, and it is not an application to be attempted once the mathematics is finished. The mechanism I expect to make it tractable is the concentration of the aerodynamic forcing into a thin layer, and the structure that concentration produces is what I intend to contribute. The reduced model of [Section 3](#) is the first test of that mechanism rather than the endpoint, and the bound is to be to hold for a stated class of vehicle models rather than for one vehicle: the physical parameters are to enter through named coefficients and explicit hypotheses on them, so that the statement is reusable and no part of it rests on calibration to a particular vehicle’s simulation data.

An entry vehicle is steered almost entirely by its bank angle. Rolling about the velocity vector turns the direction of the aerodynamic lift without changing its magnitude, so the control authority is a direction rather than a force, and the trajectory that results is a long unpowered glide interrupted by passages through denser air. A vehicle entering shallowly may leave the dense region entirely and re-enter it later; a descent of several such passes is a *skip* trajectory, and it is the case of interest here.

The design question concerns a set of trajectories rather than one. Given the vehicle, the initial state, and bounds on the atmosphere, which states can be reached at a given time, and which landing sites are consistent with the admissible controls? This set — the *reachable set* — is what a footprint, an abort analysis, or a certification argument is really about, and it is a different object from any finite collection of simulated trajectories. A simulation campaign samples the control set; a bound must hold for every admissible control,

and for every atmosphere within the stated bounds, at once. The two tasks are not the same, and the second is an analytical one. I propose to study that set in the limit in which the atmosphere is thin.

2 What makes the limit singular

Let ε denote the ratio of the atmosphere's scale height to the planetary radius. For the Earth $\varepsilon \approx 1.3 \times 10^{-3}$, and for an exponential atmosphere the density falls by a factor e over a height ε in units of the radius. The aerodynamic force is therefore appreciable only in a thin band of altitudes, and a vehicle above that band is in free flight.

Consider a vehicle that descends into the band and leaves it again. The time spent in the band vanishes with ε . Whether the change in the flight-path angle does, and in which variables it does not, is the first thing the analysis has to establish rather than assume. A force of order one acting for a time of order ε delivers an impulse of order ε , which vanishes; so an effect that survives the limit has to come either from a different force amplitude or from measuring the change against an entry angle that is itself small with ε , as the Chapman variables below would suggest. Settling that is the first question of [Section 3](#), and a crossing being brief and decisive at once — in whichever variables make it so — is the feature I propose to analyse.

Three consequences shape the choice of method.

Time-parametrised limits lose the crossings. If one passes to a limit in which trajectories are compared as functions of time, or in which the occupation measures of the family are allowed to converge weakly, the crossings occupy intervals whose lengths tend to zero and so contribute nothing to the limit, while the changes they produce persist. The limiting object one obtains in this way does not describe the family. The difficulty is not a failure of compactness — the states remain in a compact corridor — but a failure of the limit to retain a structure that the family has.

Averaging gives the wrong kind of statement. The natural first move for a system with a fast and a slow part is an averaging theorem, and the classical theory [\[1\]](#) does apply to the slow descent away from the band. It gives an order of approximation on a bounded interval for a given initial condition, and near a resonance it holds outside a set of small measure. A reachable-set bound cannot discard a set of small measure, and it requires an inclusion valid for every admissible control rather than an order of approximation for one of them. I expect averaging to be useful within the argument without supplying its form.

The right object is a differential inclusion. Because the control and the atmosphere are both to be quantified over, the family is naturally a family of differential inclusions [\[2\]](#) rather than of controlled trajectories, and the limit to be taken is a limit of their reachable sets. The uncertainty in the atmosphere enters as an enclosure — a bounded family of admissible density profiles — rather than as a single field, which is both what the application supplies and what the analysis will have to tolerate.

What I would like to arrive at is a limit object in which a crossing has become an instantaneous transition, set-valued because the control is free during a crossing, together with a one-sided bound: the reachable set of the family contained in that of the limit object, widened by a residual explicit in the data. The parameter that governs how a crossing resolves is not ε alone but the ratio $\gamma_e/\sqrt{\varepsilon}$ formed with the entry flight-path angle, as in Chapman's analysis of entry in these variables [\[3\]](#); for the Earth $\sqrt{\varepsilon} \approx 0.036$ radians, about two degrees, while entry angles are a few degrees, so the physically interesting case is the one in which that ratio is of order one. Whether a useful bound can be had there, and not only in the asymptotic regime, is one of the questions below.

3 A reduced model

Before any reduction the object itself can be named, and naming it does not presuppose the scaling that makes it attainable. The state is $x = (z, v, s)$ confined to a closed corridor X ; the control is a measurable function c with values in a compact set U , piecewise constant on a prescribed grid wherever the guidance is sampled; the atmosphere is a profile ranging over an enclosure $\mathcal{P}$, a bounded family of positive decreasing functions subject to conditions to be fixed; and the data is an initial set X_0 with a horizon T or a terminal condition. The reachable set $\mathcal{R}_\varepsilon(T)$ is the set of states attained at T over every admissible control, every profile in $\mathcal{P}$ and every initial state in X_0 . What I want is a limit object with reachable set $\mathcal{R}_0(T)$ and an explicit $\delta(\varepsilon)$ for which

$$\mathcal{R}_\varepsilon(T) \subseteq \mathcal{R}_0(T) + B_{\delta(\varepsilon)},$$

the inclusion one-sided, the ball taken in a metric on X still to be chosen, and δ a function of the data alone. Which of X , U , $\mathcal{P}$ and that metric the estimates genuinely constrain is part of the work.

I propose to do the analysis first on the smallest system in which the phenomenon above is present. Let z be an altitude above a reference level, v its rate, and s a vector of slow variables of the descent — speed, range, heading. Let ρ be a positive, decreasing density profile and write $\rho_\varepsilon(z) = \rho(z/\varepsilon)$ for the profile read on the scale ε , so that ρ_ε is of order one where z is of order ε and decays quickly above; the subscript is a label and not a power. Let c be the control and κ the drag. Both aerodynamic terms carry the factor $\rho_\varepsilon(z)$ explicitly, since neither lift nor drag acts outside the air. That is what makes the dissipated energy an integral of the density along the trajectory, which the plan below relies on. The model is

$$\dot{z} = v, \quad \dot{v} = c \rho_\varepsilon(z) - \kappa(s, c) \rho_\varepsilon(z) v - 1, \quad \dot{s} = F(s, c) + \rho_\varepsilon(z) \Gamma(s, c), \quad (3.1)$$

a particle in a potential with one soft wall and uniform gravity above it, the control entering as the strength of the wall, and a slow variable driven both by a drift and by the density. One altitude in Eq. (3.1) is not chosen but determined: the level at which $c \rho_\varepsilon = 1$, where the aerodynamic acceleration balances gravity exactly. It is the minimum of the potential, and the three regimes of entry flight are three energies of the oscillation about it — the equilibrium is the glide, a small oscillation is the phugoid, and a large one is a skip.

What the model keeps is the feature of Section 2: a force confined to a thin band and acting for a correspondingly short time, with a control that acts only while the vehicle is inside. What it drops is everything else — the attitude and elastic degrees of freedom, the rotation of the planet, the third dimension, and any vehicle-specific aerodynamics. The point of dropping them is diagnostic rather than decisive. If the structure I expect is absent from Eq. (3.1), the thin-layer mechanism does not supply it in this formulation; if it is present, the next question is which of the omitted couplings preserve it and which destroy it. A reduced model can also fail to show an effect that one of the dropped terms creates, so on its own it settles nothing about the full dynamics.

Three questions about the model itself have to be settled before any of the analysis, and I do not regard them as settled now.

How ε should enter. There is more than one way to put a small parameter into Eq. (3.1), and they do not describe the same limit. One may measure z in units of the band's thickness, so that the band has thickness one and the oscillations are of size $1/\varepsilon$; or leave the equations of motion free of ε and put it inside the density profile, as above, so that the band is thin and the oscillations are of order one. The two differ in what a crossing contributes in the limit: in the first the oscillations are all fast and a crossing's impulse scales with ε , while in the second a crossing occupies a vanishing fraction of an oscillation and its impulse does not. Neither choice, by itself, settles the matter, and an estimate decides it: the whole aerodynamic potential the band can present is of order ε , so a vehicle the band turns around cannot arrive with a vertical rate of order one, and the impulse of a crossing has to be weighed against a rate that is itself vanishing. What the right distinguished scaling is, whether it is the one the Chapman variables of Section 2 indicate, and what reduced model it yields, is the question this proposal starts from. Four quantities have to be kept apart in any such accounting, and running them together is what makes the question look settled when it is not: the amplitude of the force, which in Eq. (3.1) is of order one; the thickness of the band, of order ε ; the potential the band can present, which is the amplitude against the thickness and so of order ε ; and the impulse a crossing delivers, which is the amplitude against the time spent inside and so depends on the rate at which the band is crossed. It is the last that has to survive the limit, and it is the only one of the four not fixed by the data alone. Every statement downstream is a statement about one limit or the other, and I do not think the reduced model can be posited in advance of it: it has to be derived.

How the crossings should be delimited. An argument of this kind needs to say when a trajectory is inside the band, and the obvious device — fix a reference altitude and call the excursions below it crossings — is not satisfactory. The atmosphere has no cutoff, so such a level is chosen rather than determined; it makes the decomposition depend on a parameter that nothing in the problem supplies; and the air above the level is not zero, so the estimate must separately pay for what a supposedly ballistic arc still collects. I would rather delimit the crossings by something intrinsic to the motion, and there are two candidates: the turning points of the oscillation, at which $v = 0$ and which are properties of the trajectory alone; and the level $c \rho_\varepsilon = 1$ at which the forces balance, which is determined by the data but moves as the control varies. Deciding between them, and rebuilding the estimates on whichever is chosen, follows immediately, since every count and every duration bound in the argument refers to it.

What class of profiles is admissible. Because the atmosphere is uncertain, ρ_ε should range over a family rather than be a single function, and the family has to be small enough that the estimates are uniform over it and large enough to contain a real atmospheric enclosure. Which conditions on the family are needed, and which are merely convenient, is part of the work. The answer should take the form of a short list of conditions on the data, checkable before any trajectory is computed.

Where the analysis then begins, and what obstructs it. With those three settled, the first analytical step is a monotone quantity. Drag dissipates energy irreversibly, and for a control held fixed the vertical motion carries a mechanical energy whose decrease along a trajectory is exactly that dissipation, which is an integral of the density. A quantity of that kind would be the natural source of the *quantitative* count of crossings that Section 5 says the classical decomposition lacks.

It does not survive a varying control unchanged, and this is the obstruction I take to be first in line. The energy depends on the control through the aerodynamic part of the potential, so when the control moves there is a control-work term in addition to the dissipation, and it carries no sign. The limit I want deliberately leaves the control free during a crossing, and in the full problem the guidance is itself part of what is quantified over, so drag alone does not supply a count: what is needed is a decrease uniform over the admissible controls. Three routes seem open — a common Lyapunov function whose decrease is uniform over the admissible class; a bound on the cumulative control work under a restriction on the variation or the switching of the control; or a different monotone quantity in which the control-work term does not appear. Establishing one of them belongs to the first milestone and is not a tool already in hand. The same caution attaches to the observation above that the band's potential is of order ε : as stated it belongs to a fixed potential, and under a varying control it needs the corresponding work bound.

4 The structure I expect, and the work it comes from

The shape I expect the family to have is a skeleton carrying the slow descent, one profile for each crossing of the band, and a remainder. That expectation is not a guess about this problem; it is the structure of the problem I have worked on.

In joint work with E. Kirr I studied multi-peak ground states of the stationary nonlinear Schrödinger equation with an external potential in the high-energy regime. The method there has three steps. A rescaling turns the high-energy limit into a family of equations in a parameter tending to zero. The concentration-compactness principle of Lions [4] then decomposes a bounded family of solutions into finitely many profiles centred at distinct points, each resolved at its own scale [5, 6], with a remainder tending to zero in the energy norm. Finally a Lyapunov–Schmidt reduction projects the problem onto the kernel of the linearised operator and turns the interaction between profiles into a finite-dimensional algebraic system for the positions of the concentration points, whose admissible solutions can then be classified.

The analogy I want to test is that a crossing of the band plays the part of a peak. It has its own scale, which here is the thickness of the band, and its own centre, which here is the time at which the crossing occurs. Two features of the analogy look favourable at first sight and both need qualification. Peaks in the NLS problem have to be shown to separate, whereas crossings occur at distinct times and might be expected to have disjoint supports once rescaled; but distinct times need not stay separated on the rescaled scale, and crossings that approach one another as $\varepsilon \rightarrow 0$ may overlap or merge into a single compound transition, so separation, clustering and compound events are questions rather than properties the dynamics supplies. And each crossing dissipates energy while the total available is finite, which suggests the count is bounded by something concrete; but a finite total does not by itself exclude arbitrarily many arbitrarily weak crossings, and a uniform count needs a lower bound on what each crossing counted contributes. A workable statement may have to count only the crossings above a threshold and leave the rest in the remainder.

Two differences are the substance of the proposal, and they are why this is not an exercise in translation.

The objects are trajectories of a controlled system, not critical points of a functional. There is no variational structure to project onto, no linearised operator whose kernel supplies the reduction, and no single family to decompose: what must be bounded is the whole set of admissible trajectories. Whether anything like the Lyapunov–Schmidt step has an analogue here — a finite-dimensional system for the times and strengths of the crossings — is an open question and, if it does, probably the most interesting part of the problem.

The decomposition would have to be quantitative. The classical argument obtains a finite profile count by contradiction: if there were more than a certain number of profiles above a given size, a conserved quantity

would be exceeded. That argument gives finiteness without a constant. A reachable-set bound is useless without the constant, so the count must come from the data in advance. This is the point at which I expect the dissipated energy of [Section 3](#) to do real work, and it is the part of the programme I expect to be both tractable and decisive.

5 What would have to be established

Four problems, each stated with the obstruction it has to overcome. They are not of equal standing. Resolving the scaling of [Section 3](#) and proving a first estimate for whatever reduced model it yields is the essential milestone, without which none of the rest is well posed; [objective 2](#) is the candidate for the principal theorem; [objectives 1](#) and [3](#) are worth pursuing in so far as the first theorem creates a need for them, and not otherwise; and [objective 4](#) is the governing objective, so that every intermediate result is to be judged by whether it survives into the full problem or states the assumptions under which it would.

Objective 1. A decomposition, with a constant

That every sequence of admissible trajectories with $\varepsilon_n \rightarrow 0$ admits, along a subsequence, a splitting into a skeleton, finitely many profiles obtained by rescaling the crossings to the thickness of the band, and a remainder small in a stated norm; and that the number of profiles is bounded in advance, in terms of the data, rather than extracted by contradiction.

The obstruction is that the statement is not yet a statement about a space. Four things need specifying before it can be proved or disproved: the ambient set of trajectory objects and the group acting on it, which here should be time translation together with the rescaling of a crossing to the band's thickness; what convergence in that set means; what orthogonality of two profiles means, which I expect to be free; and in what sense the remainder is small. For the last I would propose the share of the dissipated energy held by any window of fixed rescaled width, which is the quantity in which such a remainder ought to be small and is not small in the conserved quantity. That is so far a functional on trajectories, not yet shown to be a norm or to metrize the convergence the extraction produces.

Objective 2. A limit object, and a one-sided bound

That the reachable sets of the family converge, in a sense to be chosen, to the reachable set of a limit object in which each crossing has become an instantaneous transition; and that the containment is one-sided and explicit, so that the limit object widened by a stated residual encloses the family.

Three obstructions. The transition has to be set-valued, since the control is free while the vehicle is inside the band, so the limit object is not an ordinary hybrid automaton and its admissible mode schedules have to be said rather than derived from a guard. The comparison is between a trajectory in which a crossing takes time and one in which it takes none, so the two cannot be compared at equal times and the statement has to tolerate a shift in the time parameter. And the residual has to be assembled from per-crossing errors that compound; the error in the vertical rate and the error in the slow variables couple at each crossing, so composing the estimates over many crossings is not a telescoping argument and may itself be where the difficulty lies.

Objective 3. The companion at the level of measures

That the occupation measures of the family, supplemented by a separate transition measure supported on the graph of the set-valued map, converge to a pair satisfying the Liouville equation of the limit object. The occupation-measure formulation of Lasserre, Henrion and collaborators [7, 8] renders the dynamics a linear constraint on a measure, which is what makes the solution set of a nonconvex family tractable at all, and an outer bound follows from its dual. The hybrid form of the relaxation has been studied [9], but not for a reset generated by an inner controlled problem with an integral constraint, which is what a crossing is here. I expect this formulation to be where the limit is cleanest to state and where the concentration machinery is genuinely needed, for the structure rather than for the constants. The supplement is likely to be necessary rather than cosmetic, and for the reason given in Section 2: a crossing occupies vanishing time, so ordinary occupation mass need not retain anything about it, and the mass associated with a transition of no duration is not the same object as occupation mass over an interval. How such a measure is generated by the inner crossing problem, how it couples to the Liouville equation, and whether the pair yields the outer bound, are the questions. Whether the limit is better taken here or at the level of trajectories, and whether the two routes agree, I do not know.

Objective 4. The full entry problem

Three hypotheses separate the reduced model from entry, and each is a question about what the analysis can honestly carry. The attitude and elastic motions do not concentrate. They are persistent oscillations whose frequencies cross during a crossing of the band, and averaging them is the obvious route and the wrong one, for the reason given in Section 2: an averaging theorem through a resonance holds outside a set of small measure, which an outer bound cannot discard. They act on the trajectory only through the aerodynamic force, so it is the force that must be enclosed rather than the motions averaged. This is the part I expect to be hardest and it is on the critical path. The atmosphere is uncertain and varies in time, whereas the enclosure of Section 3 is a fixed family. Allowing the profile to vary along a trajectory — within bounds, and with the monotonicity in altitude that the physics supplies — is an enlargement of the enclosure rather than a sharpening of its constants, and it is what an application would require. The guidance is issued at finite intervals, so the control is piecewise constant rather than continuous. Whether that helps or hurts is not obvious, and settling it is part of the work: a control held constant is better behaved between switches, but a switch is a discontinuity and may fall where the motion concentrates.

6 Plan of work

Four phases in sequence, since each is what the next rests on.

First, the full problem and the reduction from it. Write down the entry dynamics with the admissible guidance laws, the atmospheric enclosure and the terminal constraints, and state the guarantee wanted of them [10]. Then identify the distinguished asymptotic regimes, determine which variables move fast and which stay slow, and settle whether a passage through the layer produces a transition that survives the limit. The reduced model is to come out of that analysis; stating it first and checking it afterwards is the error I most want to avoid, and Section 3 says why.

Second, the first estimates, in the simplest adequate model. In whatever reduced system the reduction yields, bound the crossing dynamics, quantify the transition, control the number or the cumulative effect of the passages, and obtain an explicit enclosure — every constant explicit and every hypothesis a condition on the data. I have a draft route for the model as it currently stands, and the tools in it are elementary ones: a dissipation estimate, a change of variable, Gronwall's inequality and a counting argument. I do not present it as settled, and if the reduction of the first phase changes the model then the route changes with it. What I want from a first year is to bring it to the standard at which I could defend every step at a board, and to find out which steps do not survive the attempt.

Third, the decomposition and the measures. With the reduced bound secure, attack objectives 1 and 3:

make the trajectory space and its topology precise, decide whether the decomposition buys anything the elementary route does not — the limit object itself rather than a comparison, the removal of the hand-made classification of crossings, the topology in which the remainder is small — and state the measure-level companion. If none of those survives examination, the honest conclusion is that the machinery is the right language rather than a necessary tool. That conclusion would itself say something about the method, and I would state it as such.

Fourth, back to the full problem. Reintroduce what the reduction dropped — attitude, the time-varying atmosphere, sampled guidance — and quantify what each does to the transition and to the enclosure. State in each case which effect can be enclosed, which needs a stronger hypothesis, and which obstructs the guarantee altogether. This is the phase the thesis is for, and the preceding three are judged by how much of the full problem they reach.

7 Terminology

Four words are used in a fixed sense throughout, and each is borrowed from one of the two subjects. A *crossing* is an episode in which the trajectory is inside the band; in the aerospace literature this is a pass, an entry, or a skip. The *skeleton* is the slow motion that remains when the crossings are contracted to instants; it corresponds to the equilibrium glide. A *profile* is a crossing rescaled to the thickness of the band, the object that would appear in a concentration-compactness decomposition; where the distinction matters I call the corresponding piece of the limit object a *jump*. The *residual* is the difference between the family and the limit object, measured as a widening of the reachable set, and not the remainder of a decomposition, which is a different object. Where the two subjects use one word for two things — as with the *amplitude* of an oscillation and of a mode — I give the analytical sense and say so.

8 Status

So that there is no ambiguity about what exists. This is a proposal: the central statement for the entry problem, and every constant in it, are open, and none of the four problems of [Section 5](#) is solved. For the reduced model [Eq. \(3.1\)](#) my working notes set out a candidate route to the bound of [objective 2](#) in detail, with hypotheses and constants written out. They are the starting point of the first phase above and not a result I present: I do not offer them as proved, and rebuilding them is the first thing I propose to do. The published documents are this proposal and its two-page summary.

References

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